REACH

An alert isn’t finished until someone can act on it.



Voice calls in each resident’s language that confirm they understood an emergency notice, and show coordinators who needs a person.

Dual-Use Hackathon · USC Viterbi × InnovateLA · Resilience Comms lens (Retell AI) · Rafid, Enzo and Abdullah

THE PROBLEM

Emergency alerts are broadcasts. They never find out who heard them.

Assumes a smartphone

Misses people on basic phones.

Assumes English

Misses people who don’t speak it.

Never asks back

Nobody knows who didn’t hear it.

The people most likely to miss an alert are also the most likely to need help acting on it.

HOW IT WORKS

One loop: call, check, follow up

1. **An approved notice**
   A coordinator picks an approved notice and the residents to reach.

2. **A call in their language**
   A Retell voice agent reads it word for word in English or Spanish, then asks: can you follow this, or do you need help?

3. **Needs, read back**
   Transport, mobility, a call back, confirmed on the call. It never promises help is coming.

4. **A text if the call fails**
   Numbered replies: 1 = I can follow it, 2 = I need help. Then everyone lands in a dashboard lane.

Simulated evacuation warning	Call in Spanish, shown in English
REACH	...if you can't leave on your own, tell us now. Can you follow these instructions, or do you need assistance? ...si no puede salir por su cuenta, díganoslo ahora. ¿Puede seguir estas instrucciones, o necesita ayuda?
Resident	**"I understand, but I don't have a car, and my mother uses a wheelchair."** Entiendo, pero no tengo carro, y mi mamá usa silla de ruedas.
REACH	Let me read that back: you need transportation, and someone at home uses a wheelchair. Is that right? Le confirmo: necesita transporte, y alguien en su hogar usa silla de ruedas. ¿Es correcto?
Coordinator	**Test Resident B needs help** Needs transportation, wheelchair user in household, needs confirmed on the call

LIVE DEMO

90 seconds, three residents

1. Send the evacuation warning to 3 residents.

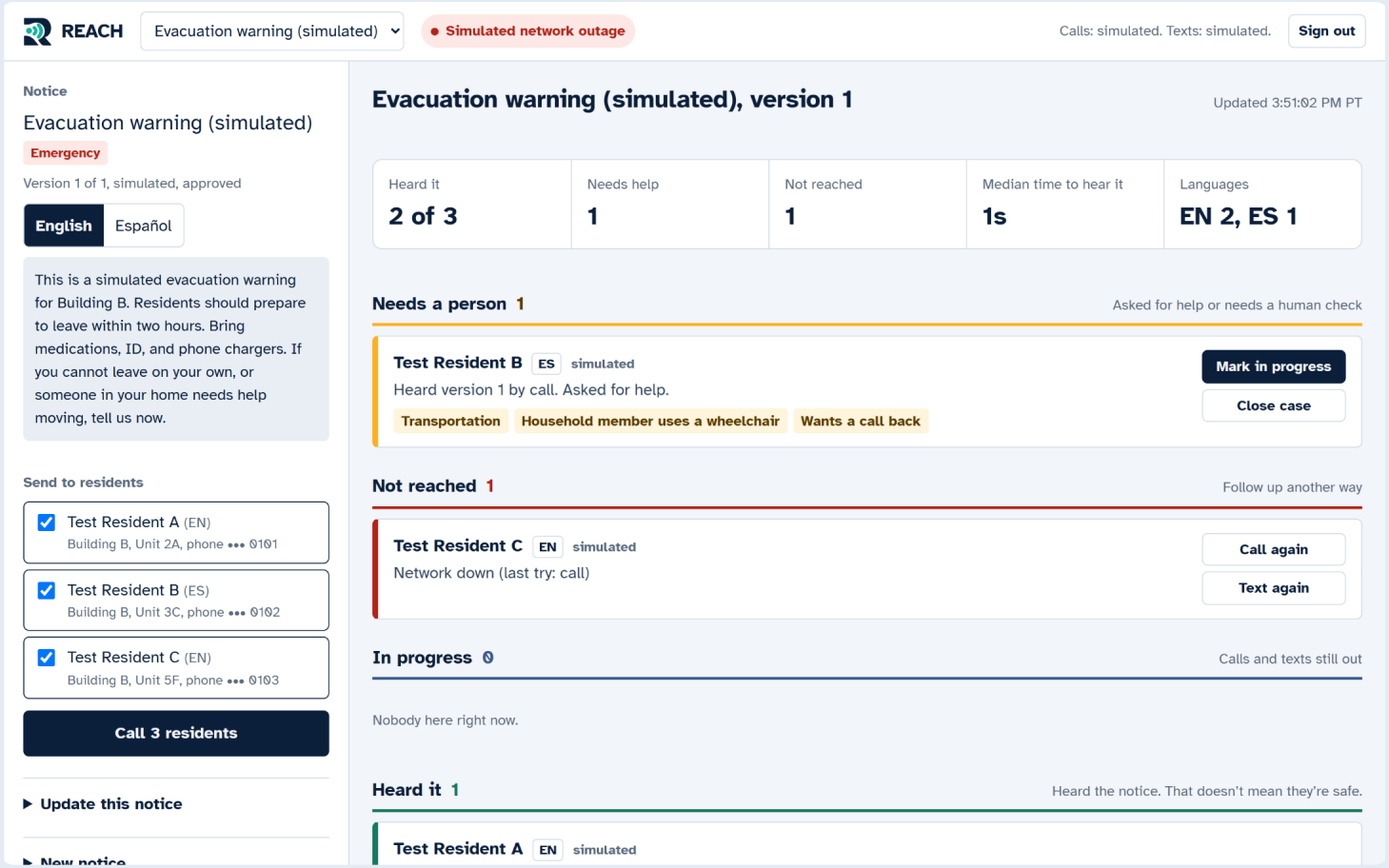
2. A confirms. B asks for help, in Spanish.

3. C doesn't answer, so REACH texts.

4. Network outage: C stays on the board as "Network down."

5. Print the door-knock list.

6. Update to version 2 and see who heard which.



THE HACKATHON

Three twists, absorbed overnight

Fri 11:31 PM

Funding

NSF SBIR/STTR Phase I with Cal Poly Pomona, then city and county buyers on FEMA UASI funds. We built the impact report they'd ask for.

Sat 4:30 AM

Go global

One region config for languages, timezone, alert authority and consent. Example: cyclone warnings in Odisha, India, with a Hindi agent.

Sat 7:00 AM

Poster

A poster for the Devpost submission, alongside the deck and demo video.

Heard it	Needs help	Not reached	Median time to hear it	Languages
2 of 3	**1**	**1**	**1s**	**EN 2, ES 1**

RESPONSIBLE BY DESIGN

Guardrails, and what’s next

Guardrails we built

- Only approved notices can be sent
- The agent says it’s automated and never promises help
- Opt-outs are honored immediately
- Webhook signatures are verified

What’s next

- A real deployment with a housing operator
- More languages
- Official alert feeds as the source
- Assigning in-person follow-ups to coordinators

Thank you · Rafid, Enzo and Abdullah · github.com/siddiquiabdullah526-wq/REACH